



ECU

Egyptian Chinese university

Selected Topic

Project: Image Classification Traditional CNN
vs Dilated CNN



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1. Introduction

Deep learning has become one of the most important technologies in computer vision and image processing. Convolutional Neural Networks (CNNs) are widely used for image classification because they can automatically learn important visual features from images.

This project focuses on comparing two different CNN approaches:

- Standard Convolution
- Dilated Convolution

The comparison is performed using two famous datasets:

- CIFAR-10
- STL-10

The purpose of the project is to understand how changing the convolution type affects image classification performance and feature extraction.

2. Project Objective

The main objectives of this project are:

1. Build CNN models for image classification.
2. Compare Standard and Dilated Convolution.
3. Evaluate model accuracy on different datasets.
4. Analyze the effect of receptive fields on learning.

The project aims to study model behavior, not only achieve high accuracy.



3. CNN Overview

CNNs are deep learning models designed especially for image-related tasks. They are effective because they can preserve spatial information and detect patterns such as edges, textures, and object shapes.

A CNN usually contains:

- Convolution layers
- Activation functions
- Pooling layers
- Fully connected layers
- Output layer

The convolution layer is the most important part because it extracts image features using filters.

4. Standard Convolution

Standard convolution uses neighboring pixels to detect local patterns in the image.

Advantages:

- Simple and efficient
- Strong local feature extraction
- Commonly used in CNNs

Disadvantages:

- Limited receptive field
- Needs deeper networks to capture larger context



5. Dilated Convolution

Dilated convolution introduces spaces between kernel elements, allowing the filter to cover a larger area without increasing the number of parameters.

Advantages:

- Larger receptive field
- Better contextual understanding
- Captures wider image information

Disadvantages:

- May lose fine details
- Sometimes it is harder to optimize

6. Datasets Used

1. CIFAR-10

CIFAR-10 contains:

- 60,000 color images
- 10 classes
- Image size: 32×32

The dataset is commonly used for image classification experiments.

2. STL-10

STL-10 contains larger and more challenging images than CIFAR-10.

It is useful for testing model generalization and performance on higher-resolution images.

Training Process

The models were trained using the Adam optimizer and evaluated using accuracy and loss metrics. The notebook also visualizes training curves and compares model efficiency.



Figure 1: Sample images from the datasets used in training.

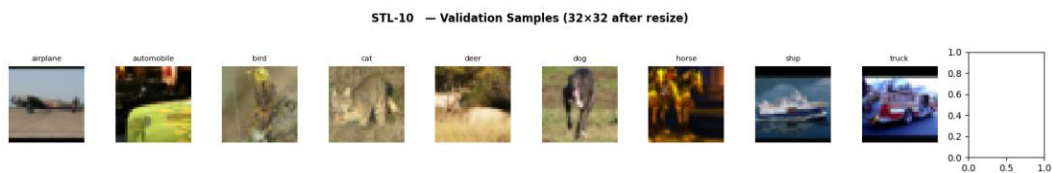


Figure 2: Class distribution visualization.

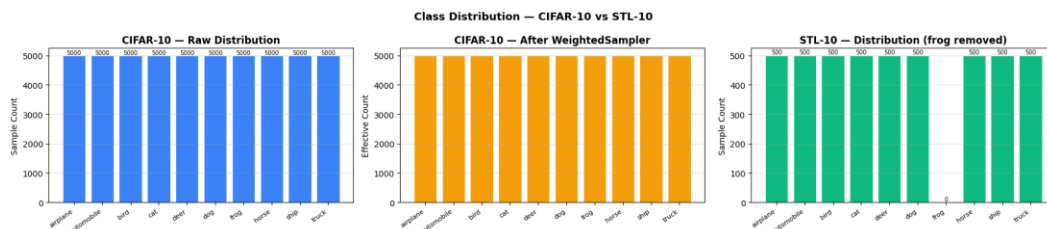


Figure 3: Learning curves showing training and validation accuracy.

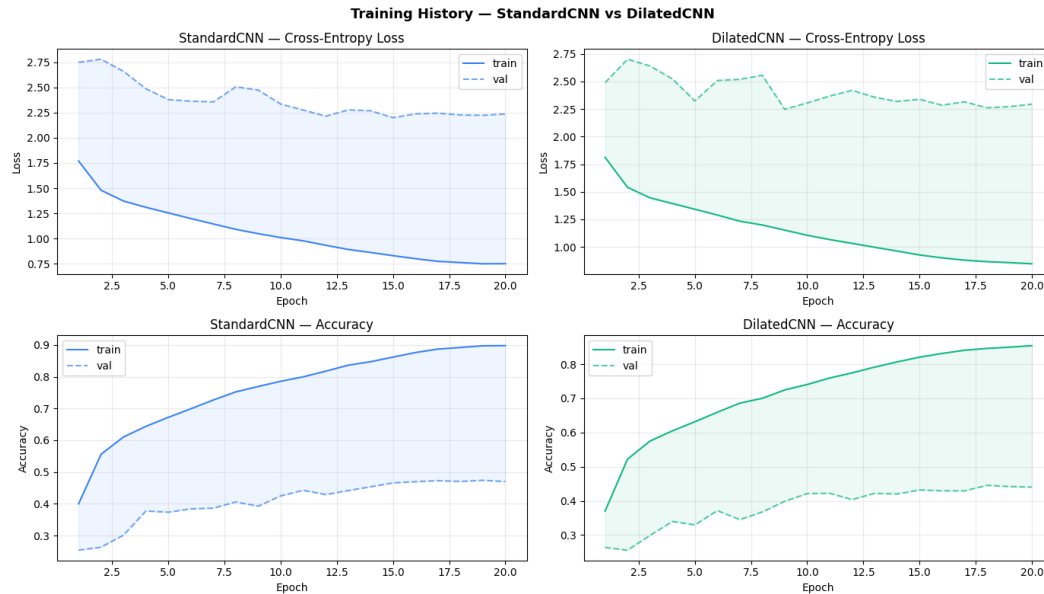


Figure 4: Performance comparison between Standard and Dilated CNN models.

7. Project Workflow

1. Importing Libraries

The project uses libraries such as:

- TensorFlow
- Keras
- NumPy
- Matplotlib

These libraries are used for building models, preprocessing data, and visualization.



2. Data Preprocessing

Before training, preprocessing is applied.

The main preprocessing step is normalization, where pixel values are divided by 255 to improve training stability.

The images and labels are also formatted correctly for CNN input.

3. Building the Models

Two CNN architectures are built:

Standard CNN: Uses regular convolution layers for local feature extraction.

Dilated CNN: Uses dilated convolution layers to increase the receptive field and capture wider contextual information.

Both models use pooling and dense layers before the final classification layer.

4. Model Training

The models are trained using:

- Adam optimizer
- Sparse categorical cross entropy loss
- Accuracy evaluation metric

During training, the notebook records:

- Training accuracy
- Validation accuracy
- Training loss
- Validation loss

These values help evaluate model learning.



5. Evaluation and Visualization

After training, the models are tested using unseen data.

Accuracy and loss graphs are plotted to analyze:

- learning performance
- overfitting
- convergence behavior

Visualization helps compare both architectures clearly.

8. Results Discussion

The comparison between Standard and Dilated Convolution shows that architecture design affects CNN performance.

Standard convolution performs well in capturing local image details, while dilated convolution captures wider contextual information using larger receptive fields.

The final performance depends on:

- dataset complexity
- network design
- hyperparameters
- image resolution

Testing on both CIFAR-10 and STL-10 helps evaluate model generalization.



9. Challenges

Some challenges faced during the project include:

- Long training time
- Overfitting
- Hyperparameter tuning
- Accuracy fluctuations

These problems are common in deep learning projects and can be improved using optimization techniques.

10. Possible Improvements

The project can be improved in the future by:

- Using deeper CNN architectures
- Applying data augmentation
- Adding dropout and batch normalization
- Testing different dilation rates
- Using transfer learning

These techniques may improve model stability and accuracy.

11. Conclusion

This project compares Standard Convolution and Dilated Convolution in CNN image classification tasks.

The results demonstrate how convolution type affects feature extraction and model performance.

Standard convolution focuses on local image details, while dilated convolution expands the receptive field and captures broader image context.



Using CIFAR-10 and STL-10 datasets provides practical experience in:

- CNN implementation
- Image preprocessing
- Model training
- Performance evaluation
- Deep learning analysis

Overall, the project provides a good understanding of CNN architecture and their effect on computer vision tasks.

12. References

1. TensorFlow Documentation
2. Keras Documentation
3. CIFAR-10 Dataset Documentation
4. STL-10 Dataset Documentation

13. GitHub Link

<https://github.com/Y0usefMohamed/Image-Classification-Traditional-CNN-vs-Dilated-CNN.git>